WORKSHEET

Chapter 6: Total change

1 The rate of flow of water (measured in L/sec) out of a dam is given by:

$f(t) = 30 + 2t$

where t is measured in seconds.

What is the total volume of water that flowed out of the dam in the first 2 minutes?

2 The rate of change in the temperature (measured in °C/min) of tea in a cup is given by:

$f(t) = 8e^{-0.2t}$

where t is measured in minutes.

What is the total change in the temperature of the tea from $t = 1$ to $t = 5$?

3 The birth rate, in thousands per year, for a population of mammals observed over a 10-year period is given by:

$b(t) = 5 + 2t$ for $0 \leq t \leq 10$

where t is measured in years.

a How many mammals are born in this period of time?

b Find the time (in years) when the number of births reaches 14 000.

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4 Liquid is leaking out of a tank at a rate given by:

$r(t) = 4000e^{-0.3t}$ litres per day and t is measured in days

a How much liquid leaks out of the tank in the first five days?

b How much liquid leaks out of the tank in the next five days?

c Why are the answers to parts **a** and **b** different?

d If the leak is never fixed, how much liquid will be lost from the tank?

5 A tank that holds a total of 5000 L of liquid completely drains in 40 minutes. The volume of liquid remaining in the tank is given by:

$$v(t) = 5000\left(1 - \frac{t}{40}\right)^2$$

a At what rate is the liquid draining from the tank at $t = 5$?

b What is the change in volume of the liquid in the tank from $t = 10$ to $t = 30$?

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6 The marginal cost for x units of a product is given by:

$C'(x) = 0.006x^2 - 1.5x + 8$ \$/unit

The fixed cost of production is $C(0) = 500\,000$ (dollars). Find the total cost of producing the first 2000 units.

7 The marginal revenue from selling x units of a product is given by:

$R'(x) = 90 - 0.02x$ (dollars)

The revenue from the sale of the first 100 units is \$8800. Find the revenue from the sale of the first 200 units.

8 The cost in dollars for producing x metres of nylon rope is given by:

$C(x) = 2000 + 3x + 0.01x^2 + 0.002x^3$ (dollars)

a Find the cost of producing the 101st metre of rope.

b Find the marginal cost function.

c What is the total change in cost when increasing production from 100 m to 300 m of rope?

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Answers

1 18 000 L

2 18.03°C

3 **a** 150 000

b 2 years

4 **a** About 10 358 L

b About 2311 L

c The rate of leakage reduces over time.

d About 13 333 L

5 **a** About 219 L/min

b 2500 L

6 $13 516 000

7 $17 500

8 **a** About $65.61

b $C'(x) = 3 + 0.02x + 0.006x^2$

c $53 400

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